



**IMS Engineering College**  
**NH-09, Adhyatik Nagar, Near Dasna, Distt. Ghaziabad, U.P.**  
Tel: (0120) 4940000

**Department: Computer Science & Engineering**

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**Object Oriented Programming with Java (BCS-403)**  
**(2023-2024) Even Semester**

**Assignment 1**

Maximum Marks: 20

Submission Date: 28-04-2025

**COURSE OUTCOMES**

By the end of the course, students will be able to:

| CO No. | Description                                                                        | Bloom's Level |
|--------|------------------------------------------------------------------------------------|---------------|
| C231.1 | Develop the object-oriented programming concepts using Java                        | K3, K4        |
| C231.2 | Implement exception handling, file handling, and multi-threading in Java           | K2, K4        |
| C231.3 | Apply new java features to build java programs.                                    | K3            |
| C231.4 | Analyse java programs with Collection Framework                                    | K4            |
| C231.5 | Test web and RESTful Web Services with Spring Boot using Spring Framework concepts | K5            |

**ATTEMPT ALL QUESTIONS each question carries equal marks**

| Q. No. | Question                                                                                                               | CO     |
|--------|------------------------------------------------------------------------------------------------------------------------|--------|
| 1.     | Explain the major features of Object-Oriented Programming.                                                             | C231.1 |
| 2.     | Compare Java Runtime Environment and Java Virtual Machine.                                                             | C231.1 |
| 3.     | How are objects implemented in Java?                                                                                   | C231.1 |
| 4.     | What are the limitations of static members?                                                                            | C231.1 |
| 5.     | What do you mean by Constructor? How many types of constructors are available in java? Explain with suitable examples. | C231.1 |
| 6.     | Describe in detail about the types of inheritance in java.                                                             | C231.1 |
| 7.     | Which is the alternative approach to implement multiple inheritances in Java? Explain, with an example?                | C231.1 |
| 8.     | What is package? How are the packages created and accessed in java.                                                    | C231.1 |
| 9.     | Explain Polymorphism and types of polymorphism with examples                                                           | C231.1 |
| 10     | Explain Interface and abstract class with examples.                                                                    | C231.1 |



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